

ShouldDex: A Panel-Woven Textile Sensing System for Rotator Cuff Rehabilitation

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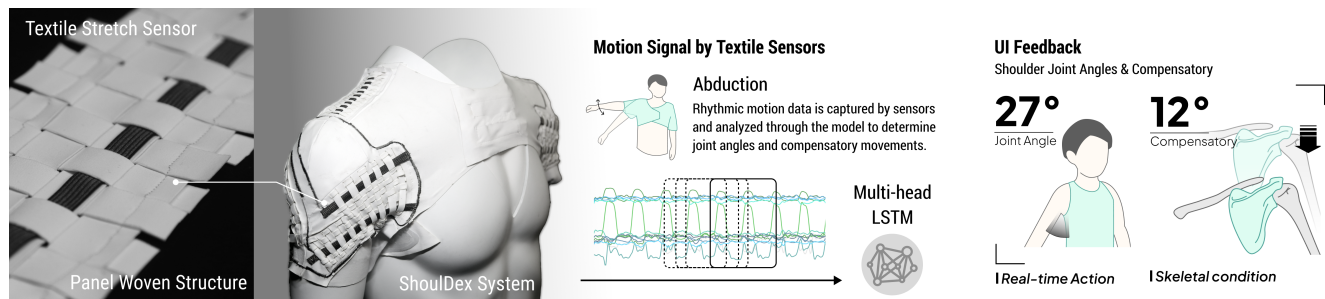


Figure 1: ShouldDex is a textile-based sensing system for rotator cuff rehabilitation. A panel-guided textile integration structure supports sensor embedding, enabling motion data collection. An LSTM model interprets the data into joint angles, which are visualized in a feedback interface to complete the training loop.

Abstract

Accurate joint angle monitoring is critical for effective rotator cuff (RC) rehabilitation. While textile-based stretch sensors provide flexibility and comfort for wearable applications, integrating them reliably into fabric-based systems remains challenging. We present a panel-guided sensor embedding approach that utilizes modified garment patterns to secure stretch sensors along predetermined pathways. Based on this method, we developed ShouldDex, a soft textile system for RC rehabilitation that integrates machine learning-driven feedback. Preliminary demonstration shows distinct signal patterns across key shoulder movements (abduction, flexion, and rotation). And a LSTM-based regression model achieves a mean

RMSE of 2.95 degrees for joint angle estimation. ShouldDex provides a promising solution to support unsupervised, home-based rehabilitation with real-time, interpretable feedback.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools.

Keywords

Textile-based Wearable System, RC rehabilitation, Shoulder Joint Angle Monitoring, Sensor Integration

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1 Introduction

Rotator cuff (RC) injuries are a leading cause of chronic shoulder pain and functional limitations [13], often requiring prolonged rehabilitation to restore joint mobility and muscular strength [20]. During resistance-based rehabilitation, accurately monitoring shoulder

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joint angles is essential for improving movement quality and tailoring personalized treatment strategies [7]. However, conventional sensing technologies such as vision-based tracking systems [11] and inertial measurement units (IMUs) [23] often struggle to capture the complex, multidimensional joint movements, especially in unsupervised or home-based training environments.

Flexible strain sensors based on knitted metallic yarns have emerged as a promising solution for RC rehabilitation monitoring [21], particularly for detecting subtle deformations that are difficult to observe with traditional methods [1, 15]. However, integrating them into garments while maintaining both signal fidelity and long-term wearability remains challenging. For example, fully stitching or adhering sensors to fabric can constrain their natural elasticity, reducing sensitivity and altering signal characteristics [8]. In contrast, anchoring only the ends of the sensors may lead to mechanical instability, wrinkling, or even short-circuiting—compromising both aesthetics and data reliability [14].

To address these challenges, we introduce a panel-guided textile integration method, embedding stretch sensors along guided weaving tracks directly within the garment’s pattern panels. This method preserves fabric flexibility, ensures precise sensor alignment, and enhances strain responsiveness during shoulder movement—while maintaining an aesthetically pleasing design. Built upon this method, we developed **ShouldDex**, a fully soft, textile-based wearable system for RC rehabilitation. Guided by biomechanical analysis, 14 textile strain sensors are strategically placed on high-deformation zones to capture both scapular and humeral motion. Preliminary tests across multiple rehabilitation movements show consistent, interpretable sensor responses suitable for multi-DOF joint angle estimation. Using an LSTM neural model, we achieved an average RMSE of 2.95° on the test set, demonstrating strong potential for real-time angle prediction.

This work contributes: (1) A panel-guided sensor integration strategy that improves signal quality, preserves elasticity, and supports aesthetic garment design. (2) A fully wearable, soft textile system that enables accurate tracking of multi-DOF shoulder movements during everyday RC rehabilitation.

2 Related Work

In rotator cuff injury rehabilitation training, accurate joint angle monitoring is of great significance for evaluating rehabilitation progress and developing personalized training programs [2]. Current shoulder joint motion monitoring utilizes systems such as computer vision [18], optical motion capture [12], and IMU sensors [15]. While these technologies can achieve high-precision joint angle measurements, they generally face the challenge of balancing accuracy and practicality, with limitations including environmental constraints, insufficient portability, and operational complexity [3, 12], making them difficult to meet the requirements of daily rehabilitation training.

In contrast, textile-based stretch sensors provide a lightweight and highly integrated solution for motion monitoring, demonstrating great potential in daily rehabilitation training monitoring [6], with applications in rehabilitation monitoring of joints such as the knee [5, 16] and elbow [4]. However, existing fabric sensor research primarily targets single-degree-of-freedom joint movements. When

it comes to the shoulder joint, a complex multi-degree-of-freedom joint, existing textile sensing solutions often combine with sensors such as IMUs [9, 10]. While this hybrid approach provides more comprehensive motion data, it also introduces challenges related to low fusion efficiency and synchronization limitations between multiple sensors [17].

To overcome these limitations and further enhance the comfort and integration of sensing systems, fully textile-based integration strategies have gradually gained attention. In practical textile sensing system applications, existing research primarily adopts full stitching or adhesive integration of sensors with garments [8], as well as integration methods that fix both ends of sensors [9, 14]. However, these approaches may restrict stretching freedom or cause structural instability and even short circuits, resulting in limitations in data reliability and motion monitoring performance.

3 Design Execution

To address the challenges of shoulder rehabilitation monitoring, we developed an integrated wearable system that combines a textile-based sensing garment with a machine learning-driven feedback module. Together, these components form a closed-loop framework for real-time motion tracking, personalized guidance, and continuous progress evaluation.

3.1 Motion Analysis for Sensor Layout

Sensor placement was guided by a biomechanical analysis of the shoulder complex, which involves coordinated motion across the sternoclavicular, acromioclavicular, and scapulothoracic joints [22]. The shoulder exhibits multiple degrees of freedom—primarily abduction/adduction, flexion/extension, and internal/external rotation—enabled by key muscle groups including the deltoid, trapezius, pectoralis major, and rotator cuff. Focusing on functional rehabilitation, we identified three representative movement patterns essential to rotator cuff recovery: abduction in the coronal plane, flexion in the sagittal plane, and axial rotation of the humerus. These movements informed the design of a sensor layout capable of capturing both large-scale and subtle motion.

A total of 14 stretch sensors were symmetrically distributed across both shoulders, as shown in Figure 2. Sensors S14 and S9, positioned near the acromion, capture the motion of the clavicle and trapezius muscles. Sensors along the anterior deltoid and pectoralis major (S10, S11, S12, S13), as well as the posterior deltoid (S1, S2, S7, S8), monitor large angular transitions and antagonistic muscle activity. Additional sensors located along the medial (S4, S5) and inferior (S3, S6) borders of the scapula detect subtle scapular gliding. This sensor layout enables comprehensive tracking of shoulder kinematics throughout rehabilitation exercises.

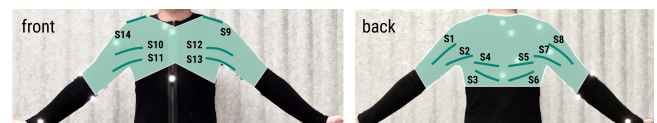


Figure 2: Sensor layout of the 14 textile stretch sensors on the ShouldDex sensing garment

3.2 Garment System Design

Based on the layout described above, the stretch sensors are integrated with the textile substrate in a way that preserves their mechanical responsiveness. To this end, we developed a panel-based woven integration method, as illustrated in Figure 3 (a).

This structure consists of three functional layers: the substrate panel at the bottom, the woven cover panel at the top, and the textile stretch sensor sandwiched in between. The substrate panel serves as the base layer, providing insulation from the skin and sweat to protect the sensor, while also offering structural support for the upper layers. The woven cover panel is aligned above the sensor path and features regularly spaced "rail-like" openings that run perpendicular to the sensor layout. These openings securely constrain the sensor's position while allowing it to slide freely within the intended direction of stretch. The stretch sensor is placed between these layers, with its endpoints fixed to the substrate using conductive threads. Its central section is loosely interlaced with the woven cover panel to allow deformation during movement.

The woven cover panels are configured differently across body regions: standard weaving for the pectoralis major and deltoid provides flatness and high fabrication efficiency, while braided structures near the acromion and scapula enable better conformity to these complex regions.

This multilayer integration ensures the sensor is tightly anchored to the garment while retaining its stretchability, allowing the sensor to deform synchronously with muscle contractions for stronger signal responses. Furthermore, the woven structure mitigates issues such as folding or wrinkling during sensor recovery, thereby preventing short circuits and improving measurement stability.

Beyond functionality, the design improves comfort, durability, and washability. Placing sensors along muscle lines forms visible "anchors" that connect physical exertion to visual deformation cues. These visual elements support proprioceptive feedback and enhance bodily awareness, contributing to improved engagement and motivation during rehabilitation training.

3.3 Hardware Implementation

The Shoulder system integrates textile stretch sensors and an electronic control unit on a highly elastic fabric foundation, achieving a balance between wearability, data accuracy, and system stability.

The base fabric is a high-stretch knitted textile composed of 85% polyester and 15% spandex, offering excellent conformity to the skin and adaptability to body deformation, which is ideal for dynamic sensing scenarios. Shoulder employs resistive textile stretch sensors, constructed from conductive yarns interwoven with elastic substrates. These sensors are fabricated using a composite knitting process that ensures mechanical flexibility, high linearity, strong robustness, and excellent strain sensitivity. Within a strain range of 20%–60%, the sensors demonstrate a significant linear correlation between resistance change and strain [19].

To ensure reliable signal transmission, a dual-layer conductive thread configuration is adopted. Non-elastic 210D silver-coated threads serve as the primary signal carriers, while 70D fine conductive yarns are used to tack them onto the fabric with a zigzag stitch pattern, as shown Figure 3 (b). This forms a flexible guiding channel that accommodates fabric stretching while reducing stress

concentration and the risk of mechanical failure. The structure not only improves the electrical stability of the system but also maintains overall textile flexibility and user comfort.

The sensing module is controlled by an Arduino Mega Pro board, equipped with a 10-bit ADC capable of simultaneously acquiring up to 16 analog input channels. Each sensor is connected via a voltage divider circuit with a 100 Ω reference resistor, allowing real-time monitoring of resistance changes. The system samples data at 40 Hz, a frequency suitable for tracking shoulder motion in typical rehabilitation scenarios. All sensor data is transmitted wirelessly via an HC-05 Bluetooth module to a host device (e.g., laptop) for real-time processing and visualization.

For daily usability and maintenance, the control unit connects to the garment using a magnetic pin interface. This detachable design enables quick removal of electronic components for washing or repair, improving durability and long-term usability of the system.

3.4 Data Processing and Feedback

To accurately perceive and visualize shoulder joint movements, we developed a complete data processing pipeline—from sensor acquisition to real-time feedback. Raw signals from the textile stretch sensors are wirelessly transmitted to a host computer via a Bluetooth module. Prior to analysis, the signals undergo low-pass filtering to eliminate high-frequency noise and ensure clean, reliable input.

The filtered data is processed using a pretrained deep learning model for joint angle estimation. We employ a Long Short-Term Memory (LSTM) network trained on ground truth data from an optical motion capture system. This model effectively captures the temporal dependencies of multi-degree-of-freedom shoulder motions and performs continuous regression to estimate joint angles.

The estimated joint angles are visualized in real time through a front-end user interface that offers interpretable feedback on movement quality. The interface includes multiple training modes and guided exercises. During initial wear, users are prompted to perform specific calibration actions to initialize the system. During training, the interface synchronously displays key joint angle data using a combination of virtual avatars and numerical indicators. Additionally, it provides real-time visualization of scapular motion and highlights sensors with significant activation, as illustrated in Figure 4. This visual feedback allows users to identify and correct improper movements during training. The closed-loop feedback mechanism not only enhances proprioceptive awareness and user engagement but also supports adaptive self-correction, ultimately improving compliance and rehabilitation outcomes.

4 Usability Demonstration

To preliminarily evaluate the usability of the proposed wearable sensing system, we conducted a demonstration involving one male participant (23 years old, right-hand dominant, with no shoulder injuries) performing three representative shoulder movements: abduction, flexion, and internal/external rotation. As shown in Figure 5, the system successfully captured real-time signals from the stretch sensors during each motion. The resulting data exhibited clear and distinguishable patterns across movement types, with consistent and interpretable signal trends, establishing a reliable foundation for joint angle regression.

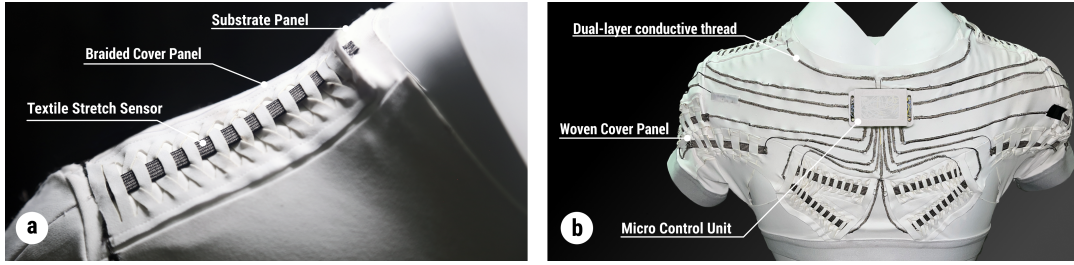


Figure 3: (a) Panel-based woven integration structure, consisting of the substrate panel at the bottom, the woven cover panel at the top, and the textile stretch sensor sandwiched in between; (b) Circuit layout and control unit of ShouldDex.

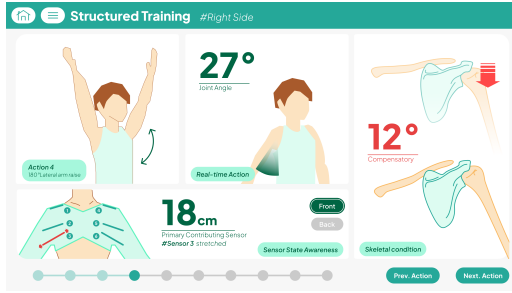


Figure 4: User Interface of ShouldDex: Displays shoulder joint angles, scapular motion, and active sensors in real time.

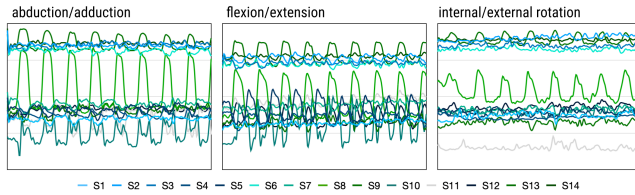


Figure 5: Textile sensor signal data during abduction, flexion, and internal/external rotation movements

For joint angle interpretation, we collected ground-truth shoulder joint angles using optical motion capture system alongside textile sensor data. We trained a multi-head LSTM network with each output head predicting a specific joint angle. On data from a single subject, the model achieved a root-mean-square error (RMSE) of 2.47 degrees on the training set and 2.96 degrees on the testing set, indicating promising accuracy for real-time estimation. These initial results demonstrate the feasibility and effectiveness of our system in capturing multi-degree-of-freedom shoulder movements and supporting future applications in rehabilitation training and motion assessment.

5 Discussion and Future Work

Sensor Integration via Panel-Based Woven Structures. We proposed a three-layer integration strategy—comprising a bottom substrate, middle stretch sensors, and a top woven guide panel—that ensures sensor alignment while allowing elastic deformation. This structure enables stable strain sensing, particularly suitable for

highly stretchable knitted sensors. Compared with stitching or adhesives, the panel-based design reduces signal noise and deformation-related errors while improving wearability. Its modularity supports adaptation to other joints. However, repeated large motions may loosen the fabric, potentially causing sensor drift over time. Additionally, the garment system requires recalibration after washing.

Cross-User Consistency and Model Robustness. Sensor signals are influenced by individual differences in fit, anatomy, and contact pressure. We used a Long Short-Term Memory (LSTM) model that learns temporal features from raw signals to predict joint angles. While effective in our single-subject test, broader generalization requires data from more users under varied conditions. Future efforts will focus on building larger, more diverse datasets and exploring domain adaptation techniques [4]. Introducing interpretable learning methods such as attention mechanism could improve both model transparency and reliability across users [21].

Real-Time Feedback for Behavior Change. ShouldDex supports rehabilitation by coupling real-time visual feedback with predicted joint angles, enabling users to promptly identify and correct movement deviations. This facilitates more accurate motor execution. The closed-loop mechanism of perception, feedback, and adjustment promotes user engagement and training effectiveness. Future developments may incorporate gamified elements or virtual coaching to support behavioral adaptation and cognitive engagement throughout the rehabilitation process.

6 Conclusion

This paper presents ShouldDex, a soft, fully textile-based wearable system for rotator cuff rehabilitation, integrating knitted stretch sensors via a panel-guided weaving architecture. The design enhances sensor stability, structural compliance, and wearability while maintaining aesthetic appeal. Preliminary tests show consistent, distinguishable sensor responses across various shoulder motions, enabling accurate angle estimation with a deep learning model. These findings highlight the system’s potential for scalable, user-friendly home-based RC rehabilitation and beyond. The source code is available on <https://github.com/Reneessense/wearable-shoulder-estimation>.

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